



Enhanced electrical properties of $(\text{Bi}_{0.5}\text{Na}_{0.5})\text{TiO}_3$ -modified BiFeO_3 - BaTiO_3 lead-free ceramics

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ARTICLE INFO

Keywords:

BiFeO_3 - BaTiO_3

Lead-free ceramics

Piezoelectric properties

Phase transition temperature

ABSTRACT

There is a significant challenge for current BiFeO_3 - BaTiO_3 based piezoelectric ceramics to achieve an enhancement in piezoelectric performance without sacrificing phase transition temperature. Here, a new lead-free ceramics material system of $(1-x)(0.7\text{BiFeO}_3-0.3\text{BaTiO}_3)-x(\text{Bi}_{0.5}\text{Na}_{0.5})\text{TiO}_3-5\text{ mol}\%\text{Bi}_2\text{O}_3$ (BF-BT- x BNT) is reported, achieving significant improvement in piezoelectric properties at stable phase transition temperature with piezoelectric constant $d_{33} = 202\text{ pC/N}$ and phase transition temperature $T_m = 484\text{ }^\circ\text{C}$, respectively. The synergy effect of reduced the ratio of rhombohedral (R) and tetragonal (T) phases, the coexistence of micro-domains and macro-domains, appropriate grain size, enhanced relative density and diminished leakage current contributes to favorable ferroelectric and piezoelectric properties. The investigation of BNT-doping on the BF-BT-based piezoceramics will provide some useful references for the development of high-performance BF-BT piezoelectric materials.

1. Introduction

As a functional material capable of attaining the interconversion of electrical properties and mechanical properties, piezoelectric ceramics possess wide applicability in piezoelectric material devices like transducers, actuators and sensors, occupying an absolute advantage in the market [1–3]. Among them, $\text{Pb}(\text{Zr,Ti})\text{O}_3$ (PZT) has been widely used in piezoelectric material devices owing to its superior piezoelectric properties and temperature reliability [4–6]. However, owing to the volatilization of toxic Pb elements in the production, use and abandonment of lead-based ceramics, there has emerged a significant potential threat to both the natural environment and public health. Hence, electronic components containing lead, specifically the applications of lead-based ceramics, have been regulated by legislation in many countries [7,8]. Therefore, the development of environmentally friendly lead-free piezoelectric ceramics to replace lead-based ceramics has become an urgent demand for social industrial production.

At present, the perovskite type (ABO_3) lead-free piezoelectric ceramics have garnered significant interest owing to their relatively superior overall electrical properties, and it can be divided into four categories: bismuth sodium titanate (BNT) based, potassium sodium niobate (KNN) based, bismuth ferrite (BFO) based, and barium titanate (BT) based ones [9]. Among them, BFO is one of promising candidate

materials because of its high Curie temperature ($T_C \sim 830\text{ }^\circ\text{C}$) and high spontaneous polarization ($P_s \sim 100\text{ }\mu\text{C}/\text{cm}^2$) [10–12]. However, pure BFO is difficult to polarize due to high leakage current and loss, resulting in unsatisfactory piezoelectric properties. According to the reports [13, 14], the formation of BF-based solid solutions is often considered an effective approach to enhancing electrical properties, making bismuth ferrite-barium titanate (BiFeO_3 - BaTiO_3 , BF-BT) one of the most extensively researched candidate materials.

For BF-BT-based ceramics, the challenge arises from the volatilization of Bi^{3+} ions and the change in the valence state of Fe^{3+} during the sintering process, which impedes the achievement of high piezoelectric properties. To overcome this and enhance the piezoelectric characteristics of BF-BT ceramics, strategies such as precise phase boundary control [15–19], process optimization [20,21] and doping elements [22, 23] were proposed. The composition of $(1-x)\text{BiFeO}_3-x\text{BaTiO}_3$ in the vicinity of the morphotropic phase boundary (MPB) will exhibit superior piezoelectric properties, because the ferroelectric domain rotates more easily near the phase boundary. Kumar et al. [19] explored in depth the relationship between ceramic composition's phase and BF content, discovering that the MPB is located near a 30 % molar content of BT, while Lee et al. [24] suggested that the MPB is at 0.33BT with a maximum d_{33} of 116 pC/N. Therefore, $0.7\text{BiFeO}_3-0.3\text{BaTiO}_3$ is the optimal composition with high residual polarization and a high d_{33}

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<https://doi.org/10.1016/j.jalcom.2024.178322>

Received 27 October 2024; Received in revised form 18 December 2024; Accepted 25 December 2024

Available online 26 December 2024

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coefficient. The process optimization of ceramics is mainly divided into synthesis method optimization and heat treatment, such as spark plasma sintering and water quenching [25,26]. Lee et al. [27] investigated BF-33BT ceramics modified with 1 mol% BiGaO₃ and subjected to water quenching, achieving remarkably high properties characterized by a d_{33} of 454 pC/N and a T_C of 454 °C, respectively, significantly promoting the development of BF-BT-based ceramics. Nevertheless, water quenching poses a serious threat to the mechanical attributes of the ceramics and has poor repeatability. So far, doping elements and compounds remain the mainstream means of modification. In recent years, some dopants, including Mn [23], Cu [28], Zn [29], Ga [30], Bi(Zn_{1/2}Ti_{1/2})O₃ [11], Bi (Li_{1/2}Ti_{1/2})O₃ [31], have been doped into BF-BT ceramics to achieve high dielectric constants or large strains. Furthermore, an excess of 5 mol% bismuth aids in compensating for the amount of volatilization during the sintering process of the ceramics, thereby improving their density and enhancing their electrical properties [32]. Therefore, the design of suitable doping schemes contributes to enhance the ferroelectric and piezoelectric properties of BF-BT ceramics.

In this work, 0.7BiFeO₃-0.3BaTiO₃ ceramics doped with (Bi_{0.5}Na_{0.5})TiO₃ (BNT) and 5 mol% Bi₂O₃ excess were fabricated via solid-state sintering to achieve higher piezoelectric constants and phase transition temperature. With the addition of a small quantity of BNT, the ceramics demonstrate a reduced R/T phase ratio and heightened octahedral distortion. The introduction of BNT and excess Bi content

serves to inhibit the valence alteration of Fe³⁺ and decrease the leakage current, resulting in high piezoelectric properties with $d_{33} = 202$ pC/N and $T_m = 484$ °C. Additionally, a systematic exploration of the relationship between the electrical properties, phase structure, and microstructure of BF-BT ceramics was conducted to understand the mechanism behind their high piezoelectric performance.

2. Experimental section

A series of (1-x)(0.7BiFeO₃-0.3BaTiO₃)- x(Bi_{0.5}Na_{0.5})TiO₃-5 mol% Bi₂O₃ (x = 0.000, 0.001, 0.002, 0.003) ceramics were prepared using the traditional solid-state reaction method. The crystal and electric properties were characterization using some analysis technologies. The detail experimental procedures were listed in [Supporting materials](#).

3. Results and discussion

Fig. 1 displays the X-ray diffraction (XRD) patterns, Rietveld-refined XRD patterns, the corresponding phase fraction and lattice constants of the BF-BT-xBNT ceramics. According to **Fig. 1(a₁)**, all ceramic samples possess stabilized perovskite crystal structures, along with a minor presence of Bi-rich impurity phases Bi₂₅FeO₃₉. This suggests that (Bi_{0.5}Na_{0.5})TiO₃ integrate into the solid solution composed primarily of the BiFeO₃-BaTiO₃ well [32]. To investigate the phase structure

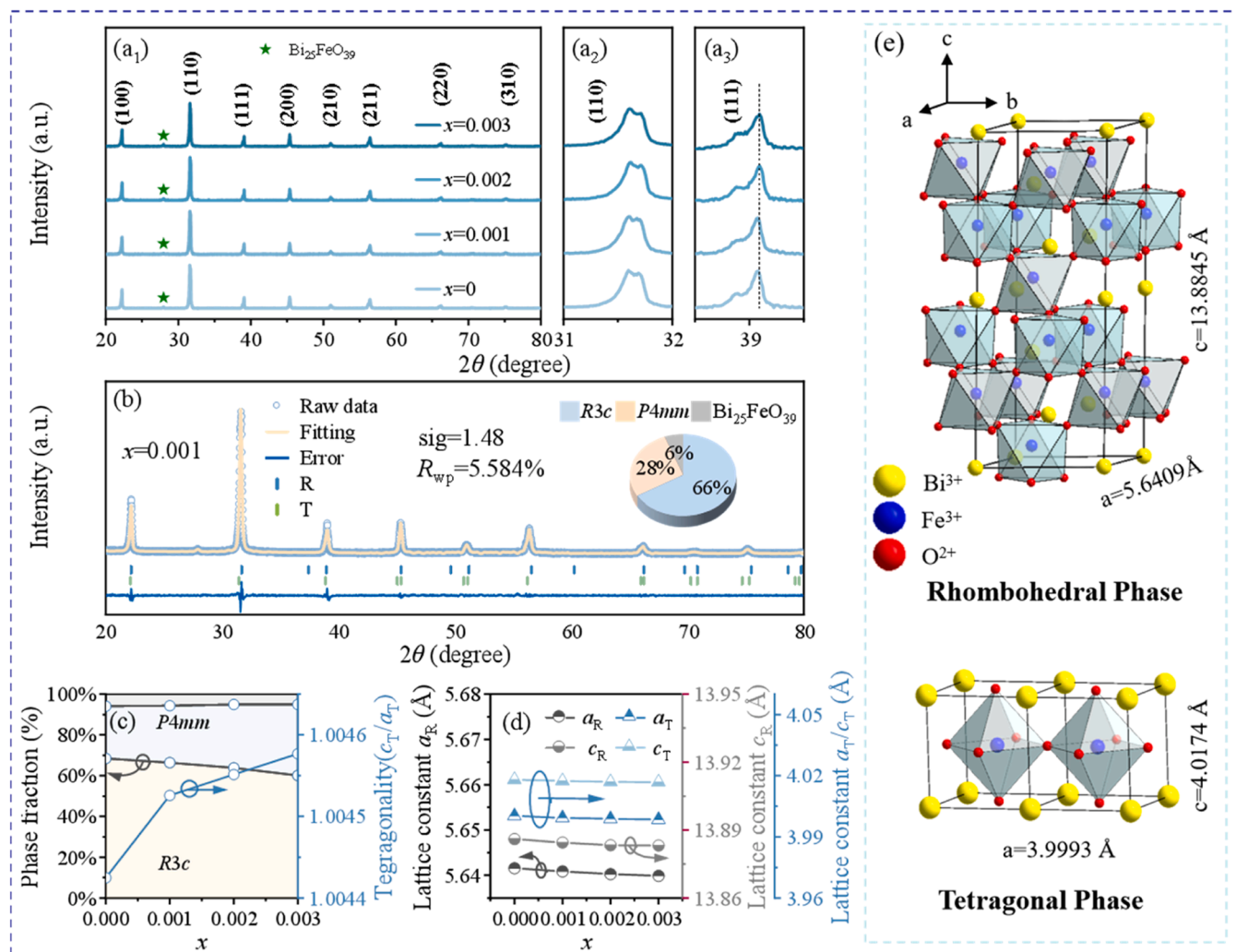


Fig. 1. XRD patterns of BF-BT-xBNT (0.000 ≤ x ≤ 0.003) ceramics are presented within the 2θ range of (a₁) 20–80°, (a₂) 31–32°, (a₃) 38.5–39.5°; (b) Rietveld refinement XRD patterns are presented within the 2θ range of 20–80° of the BF-BT-xBNT at $x = 0.001$; (c) changes in phase fraction and tetragonality as a function of x ; (d) lattice parameters of BF-BT-xBNT ceramics; (e) diagram illustrating of distortion of oxygen octahedron in rhombohedral and tetragonal phase with $x = 0.001$.

coexistence and variations in ceramics as influenced by the BNT content, Fig. 1(a₂) - (a₃) present a detailed XRD analysis focusing on the 2θ ranges of 31° - 32° and 38.5° - 39.5° , respectively. With an increase in BNT-doping content, both the (110) and (111) diffraction peaks exhibit a slight shift towards higher angles, suggesting a reduction in the unit cell volume. This shift can be explained by the replacement of the host atom Ba^{2+} ($r = 1.61 \text{ \AA}$, CN=12) and Bi^{3+} ($r = 1.03 \text{ \AA}$, CN=6)³³ by the chemical modifier $(\text{Bi}_{0.5}\text{Na}_{0.5})^{2+}$ ($r = 1.21 \text{ \AA}$ [33,34], abbreviated as BN^{2+} in the following text) in A-site, and the replacement of Fe^{3+} ($r = 0.645 \text{ \AA}$, CN=6) by Ti^{3+} ($r = 0.605 \text{ \AA}$, CN=6) [35]. The ionic size mismatch between the substituted and substituting ions will inevitably lead to lattice distortion of the BF-BT-xBNT ceramics, ultimately enhancing their piezoelectric performance [36]. Additionally, for BF-BT-based ceramics, the rhombohedral phase typically exhibits two split peaks in the 2θ range of 31° - 32° and 38.5° - 39.5° , while the tetragonal phase is characterized by a single split peak in the 2θ region around 31° - 32° . As x content rises, the pronounced splitting of the (110) diffraction peak gradually diminishes, implying the coexistence of rhombohedral (R) and tetragonal (T) phases in the ceramics [18]. When $x = 0.003$, the pronounced overlapping peaks nearly merge into a single sharp peak, marking the transition from R to T Phase. It shows the dominance of T phase on a macroscopic scale when x increases to 0.003 [18,37]. In order to gain a clearer understanding of the phase structural changes in the ceramics, we performed peak fitting on the XRD (110) and (111) peaks, and the results are shown in Supplementary Fig. S1. The peaks for compositions with $x = 0$ -0.004 were perfectly fitted to both the R phase and T phase. As x increases, the proportion of the R phase decreases while the tetragonal phase continuously increases, which is consistent with the previous conclusions.

To delve deeper into the ceramic's phase analysis, Rietveld-refined powder XRD data based on R (R3c), T ($P4mm$) and $\text{Bi}_{25}\text{FeO}_{39}$ (R3c) phases was conducted. As illustrated in Fig. 1(b) and Supplementary Fig. S2, the raw data, fitting and error curves, are provided for the BF-BT-xBNT ceramics. The XRD refinement profiles show that the refined curve fits well with the experimental data. The accuracy of structural refinement is typically assessed based on the sigma value (σ) and the Rietveld parameter (R_{wp}), which is within a credible range of 1.44-1.66 and 5.84 %-6.60 %, respectively. At normal temperature conditions, all ceramic specimens demonstrate the concurrent existence of both the R phase and the T phase, with approximately 5 % of the impurity phase $\text{Bi}_{25}\text{FeO}_{39}$ present. The phase fractions, lattice parameters and distortion of the oxygen octahedron in rhombohedral and tetragonal phases of the BF-BT-xBNT ceramics are summarized in Fig. 1(c)-(e) and Supplementary Table S1. The distortion level of the lattice can be gauged by the tetragonality ratio (c_T/a_T) of the T phase. With an increase in x content, the T phase exhibits a higher tetragonality value, which may help reduce the decline in the phase transition temperature [38]. As we can see that the lattice constants c and a of both the T phase and R phase exhibit an decreasing trend with increasing x content, consistent with the previous XRD analysis. In particular, at $x = 0.001$, lattice constants a and c decrease and tetragonal distortion increases, in which the oxygen octahedral structures of the tripartite lattice and the tetragonal lattice are shown in Fig. 1(e). The large radius of bismuth, barium and sodium atoms occupy the A-site, while the small radius of iron and titanium atoms occupy the B-site. On the one hand, the smaller-radius BN^{2+} and Ti^{4+} ions substitute Ba^{2+} at the A-site and Fe^{3+} at the B-site, respectively, causing the contraction of the cell. On the other hand, since the electronegativity of Bi^{3+} (2.02) and Na^+ (0.93) are greater than that of Ba^{2+} (0.89) [33], the BN^{2+} substituted ion has a greater attractive force to O^{2-} , causing the distortion of the oxygen octahedron [39]. With the increase of doping amount, the distortion degree of oxygen octahedron increases.

Raman spectroscopy possesses high sensitivity to alterations in the microscopic crystal structure of materials, enabling its application in studying lattice vibrations that indicate the symmetry of crystal structures [40]. To delve deeper into the local structural variations, the

Raman spectra collected at room temperature within the range of 123 - 800 cm^{-1} were decomposed into several distinct peaks in the Fig. 2 (a). According to reports, pure BFO materials belonging to the R3c space group exhibit primarily 13 Raman active modes which can be classified as $\Gamma_{R(R3c)} = (4A_1 + 9E)$ [41]. Among them, A mode corresponds to a non-degenerate state of symmetry to the principal axis, whereas E mode stands for a double degeneracy state. However, due to the presence of the weak and overlapped modes, certain specific Raman active modes are not observable or detectable in ceramic materials composed of BF-BT [42]. For BF-BT-based ceramic materials, Raman peaks are typically categorized into three regions [43]. In the region of low wavenumbers ($\leq 150 \text{ cm}^{-1}$), Raman peak modes correspond to the vibration of the A-site ion, with the patterns of A_1 -1 and E 1 being unobservable. The modes in second region (150 - 400 cm^{-1}) are concerned with the vibration of B-O bond, which principally represents the vibration of the B-site ion. As shown in Fig. 2(b), the peaks of E-2, A_1 -2 and A_1 -3 shift to a lower wavenumber region, implying a weakening of both the non-degenerate state and the doubly degenerate state of B-O bonds [44]. The shifts towards lower wavenumbers observed in the E-5, E-6, and E-9 modes indicates that the doping of BNT weakened the hybridization between the oxygen p-orbitals and the empty $\text{Fe}^{3+}/\text{Ti}^{4+}$ d-orbitals. Simultaneously, it is also intimately linked to the development of nanodomains [45]. The E-7 modes in the high wavenumber region ($\geq 400 \text{ cm}^{-1}$) is closely associated the distortion of BO_6 octahedron. In general, we can use the intensity ratio of $I(\text{E-7})/I(\text{A}_1\text{-4})$ to measure the degree of distortion of the BO_6 octahedron in different ceramic components. The ratio of $I(\text{E-7})/I(\text{A}_1\text{-4})$ in ceramic samples, as clearly depicted in Fig. 2(d), augments at $x = 0.001$, indicating that trace doping of BNT will blow up the distortion of BO_6 octahedral. Additionally, drastic change in the cell parameters may contribute to the increase of octahedral distortion. It is generally believed that the full width at half-maximum (FWHM) of the Raman band is correlated with the symmetry of BO_6 octahedron. The FWHM of some characteristic modes is summarized in Fig. 2(c). The FWHM of A_1 -2, E-2, E-6, and E-9 modes increase with x content. This broadening is attributed to several factors: the heightened structural disorder induced by BNT doping, the overlapping of phonons in distinct Brillouin zone regions, and the enhanced local heterogeneity [42]. All the above-mentioned results indicate that incorporating BNT-doping into the structure leads to disorder and enhances the random fields [41].

According to reports published [43,46], alterations in valence state of Fe ions and the presence of charged imperfections, like oxygen vacancies ($\text{V}_\text{O}^{\bullet\bullet}$), contribute to substantial leakage currents and dielectric losses, thus affecting piezoelectric and ferroelectric properties. Therefore, in order to evaluate the scenario of $\text{Fe}^{3+}(2p)$ transitions and the generation of oxygen vacancies in ceramics, X-ray photoelectron spectroscopy (XPS) analysis was conducted on BF-BT-xBNT ceramics, as illustrated in Fig. 2(e-g). For $x = 0.000$ and 0.001 samples, there are two discernible broad asymmetric peaks associated $\text{Fe } 2p_{1/2}$ with and $\text{Fe } 2p_{3/2}$ shown in Fig. 2(e). The XPS spectrum of $\text{Fe } 2p_{3/2}$ exhibits two peaks situated around 709.4 eV and 710.9 eV , demonstrating the coexistence of Fe^{2+} and Fe^{3+} . Based on the fitting results, the concentrations of Fe^{3+} and Fe^{2+} were found to be in ratios of 69.5: 30.5 and 78.7: 21.3 with increase x contents, indicating that the conversion of Fe^{3+} to Fe^{2+} in the ceramic is inhibited [as illustrated in Fig. 2(g) and summarized in Supplementary Table S1]. This may be attributed to the substitution of Fe^{3+} by Ti^{4+} . In particular, the binding energy for $x = 0.001$ sample exhibited a shift towards higher values than that of the $x = 0.000$ sample. Fig. 2(f) summarizes the asymmetrical O 1s photoelectron peaks, which are divided into three sub-peaks. For all ceramics samples, the lowest binding energy O_I peak ($\sim 529.5 \text{ eV}$) is assigned to the cation-oxygen bonds. The higher energy peaks, O_{II} ($\sim 530.3 \text{ eV}$) and O_{III} ($\sim 531.4 \text{ eV}$), are indicative of the presence of oxygen vacancies and the absorbed H_2O , respectively [47]. Compared with the undoped ceramics, when $x = 0.001$, O_I, O_{II} and O_{III} changed from 29.8 %, 26.3 % and 43.9-27 %, 23 % and 50 % respectively. It is worth mentioning that as the doped BNT-content increase, the peaks area ratio representing

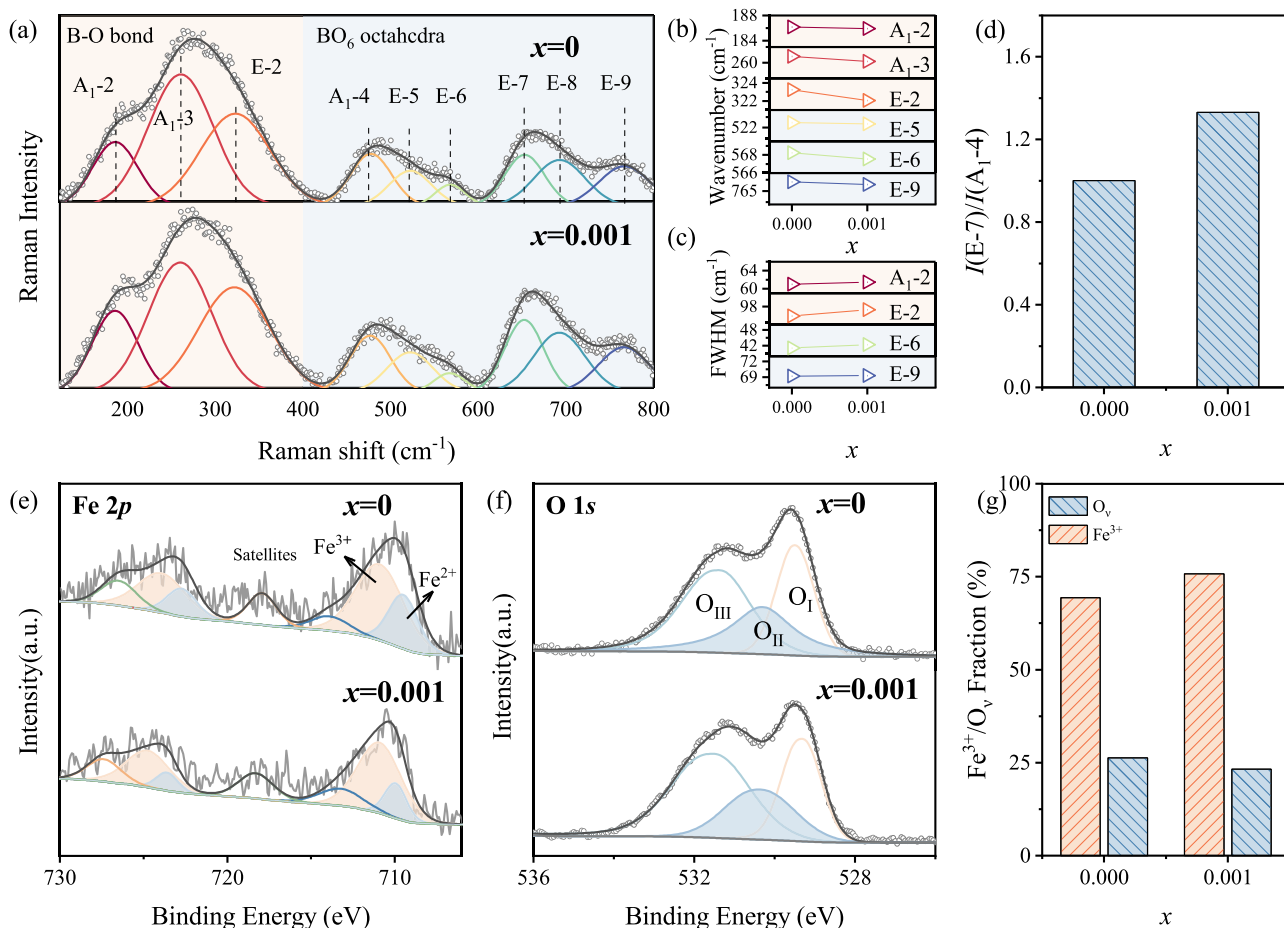


Fig. 2. (a) Composition-induced shifts in fitted Raman spectra; (b) peak positions as a function of composition; (c) variation in FWHM of various modes as a function of composition; (d) Composition-induced variations in the intensity ratio of $I(E-7)/I(A_1-4)$; Composition-induced variations in XPS spectra: (e) Fe 2p and (f) O 1s; (g) fractional abundance of Fe^{3+} ions and $V_O^{\bullet\bullet}$ ions.

oxygen vacancy reduced significantly while the area ratio of adsorbed oxygen increased, as shown in Fig. 2(g) and Supplementary Table S2. The complex ion substitutions induced by BNT doping suppress the reduction of Fe^{3+} , thereby reducing the formation of oxygen vacancies. As shown in the following Eq. (1):



It is established that oxygen vacancies have the capability to serve as the center of the pinning effect, impeding the movement of domain wall [36]. Therefore, reducing the level of oxygen vacancy can boost the efficiency of domain switching, which in turn improving the piezoelectric properties of the material. As aforementioned, it is founded that doping BNT can inhibit the change valence of Fe^{3+} and the generation of $V_O^{\bullet\bullet}$, thus reducing the leakage current and improving the piezoelectric properties of ceramics.

For classical ferroelectric materials, there exists a characteristic temperature at which the material transitions from the ferroelectric phase to the paraelectric phase, known as the Curie temperature (T_C). T_C is typically identified as the temperature at which the dielectric peak occurs in $\epsilon_r - T$ measurement, denoted as T_m . However, for relaxor ferroelectrics, T_m varies with frequency [48]. As the content of BNT increases, the dielectric peak of the BF-BT-xBNT ceramic broadens obviously, and T_m gradually moves to the high temperature direction with the increase of frequency. This behavior is indicative of the relaxor behavior of the ceramics. According to previous reports [49,50], in the following discussion, we use the change of T_m at 1 kHz to analyze the variation in the ceramic phase transition.

The impact of BNT content on the phase transformation of BF-BT-xBNT ceramics was investigated by measuring the temperature-dependent dielectric constant (ϵ_r) and dielectric loss ($\tan\delta$) [51], as illustrated in Fig. 3(a)–(d). It is clearly observed that all the ceramics exhibit a dielectric anomaly peak only around 480 °C within the testing range, with no significant dielectric anomalies at other temperatures. This indicates that within the test range, the BF-BT-xBNT ceramics undergo a ferroelectric-to-paraelectric phase transition near the T_m . This phenomenon is consistent with the previous conclusion regarding the coexistence of R - T phases at room temperature. As shown in Fig. 3(f), compared to the undoped ceramics, the ceramics with $x = 0.001$ show almost no decrease in T_m (484 °C). Whereas as x increases, T_m decreases to 478 °C. That's because the doped non-lone pair ion replaced the lone pair ion Bi^{3+} , thus interfering with the polar coupling in the system [36]. Moreover, the reduction of T_m is significantly closely linked to the decreased R/T phase ratio [34], which aligns with the XRD results. The $\tan\delta$ stays consistent at low temperatures and then quickly escalates as x increase. Additionally, the introduction of BNT content leads to a gradually broadened dielectric curve, implying the phase transition became increasingly diffused. The atomic-scale heterogeneous composition resulting from the increased A-site occupancy by dopant ions leads to relaxation behavior. The difference in dielectric behavior can be traced to the comparison between the lattice-driven intrinsic effect and the domain-induced extrinsic effect [52]. To gain a deeper understanding and quantitative assessment the diffusion and relaxation characteristics of phase transitions, we employ a revised Curie-Weiss law to represent the degree of phase transition diffusiveness, as shown by the Eq. (2) as following:

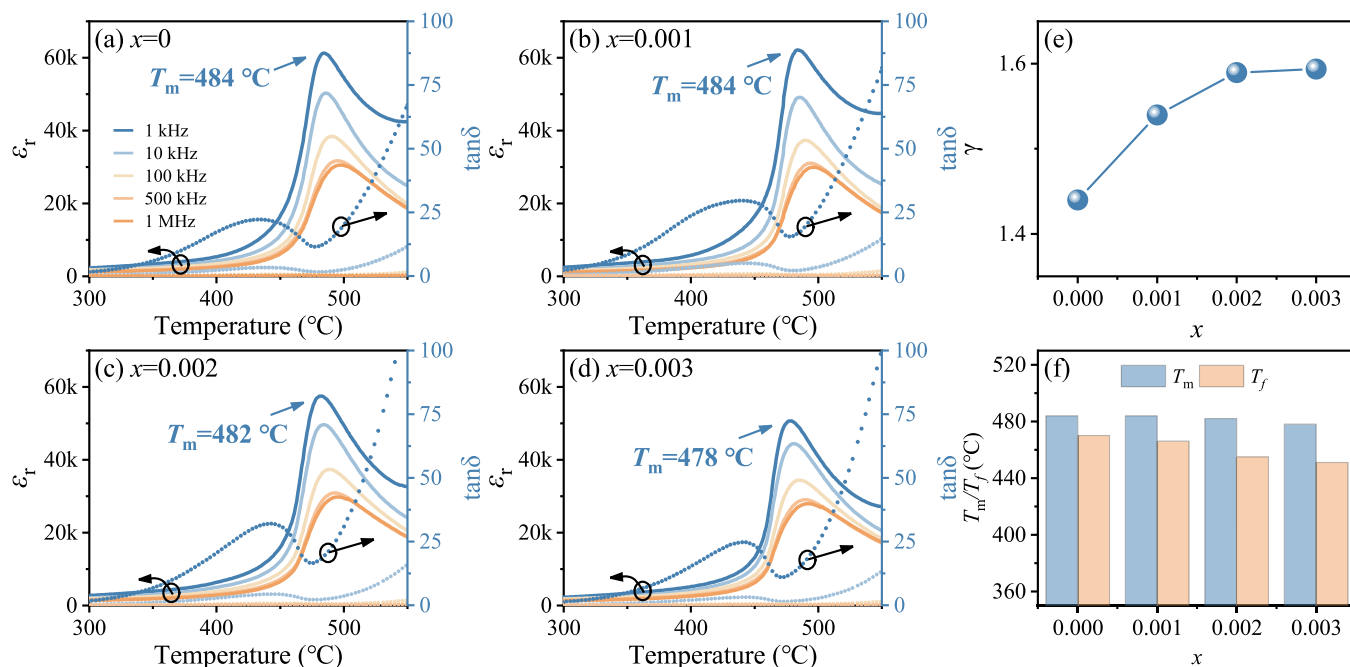


Fig. 3. (a–d) Temperature-dependent dielectric constant ϵ_r and dielectric loss $\tan\delta$ of BF-BT-xBNT ceramics with $x = 0.000, 0.001, 0.002, 0.003$ measured at 1 k, 10 k, 100 k, 500 k and 1 MHz in the temperature range of 300–550 °C; (e) diffuseness factor (γ) plotted between $\ln(1/\epsilon_r - 1/\epsilon_m)$ and $\ln(T - T_m)$ with the linear fitting at 10 kHz; (f) composition dependence of freezing temperature T_f and T_m ($f=1$ kHz).

$$1/\epsilon_r - 1/\epsilon_m = (T - T_m)^\gamma / C \quad (2)$$

where ϵ_r is the dielectric constant, ϵ_m is the maximum dielectric value, T_m is the temperature corresponding to ϵ_m , C is the Curie-Weiss constant, and γ is the diffuseness coefficient. Generally, $\gamma = 1$ and $\gamma = 2$ represent normal ferroelectrics and ideal relaxor, respectively, while the values of γ between 1 and 2 indicate relaxor ferroelectrics. Fig. 3(e) and Supplementary Fig. S3 display the fitting results of the $\ln(T - T_m)$ as the independent variable versus $\ln(1/\epsilon_r - 1/\epsilon_m)$ function. The γ values increases with x , indicating the strengthened dielectric relaxation.

In addition, the relaxation behavior of the ceramics can also be characterized by fitting the experimental data to the Vogel-Fulcher law, as shown in Eq. (3):

$$f = f_0 \exp\left(-\frac{E_a}{k(T_m - T_f)}\right) \quad (3)$$

where f is the driving frequency, f_0 is the characteristic frequency, E_a is the activation energy, k is the Boltzmann constant, and T_f is the freezing temperature, representing the boundary between the ergodic and non-ergodic states [48,53]. The fitting results are shown in Fig. 3(e) and Supplementary Fig. S4. With the doping of BNT, the relaxation behavior of the ceramics improves. Meanwhile, with the variation in composition, the T_f of ceramic exhibits a slight decrease from 470 °C to 451 °C. Specifically, when $x = 0.001$, $T_f = 466$ °C. It may imply that the doped ceramics are still capable of maintaining a relatively high operational temperature limit [54].

Fig. 4(a–d) present the SEM images of the polished surfaces (a–b) and thermally etched surfaces (c–d) for BF-BT-xBNT ceramics with different amounts of BNT. When $x = 0.001$, the ceramics have a lower porosity. At the same time, Fig. 4(i) illustrates that the ceramic density increases as the doping amount increases, possibly due to a reduction in unit cell volume. Furthermore, Fig. 4(e–h) provide illustrations that depict the grain size distributions. Clearly, as component x increases, the grain size in ceramics gradually enlarges and becomes more uniform. The enhanced growth of the grains can be ascribed to the increase of impurity phase $\text{Bi}_{25}\text{FeO}_{39}$ content, which forms a liquid phase and facilitates grain growth during sintering [34]. At the same time, the

phenomenon of partial grain refinement also gradually increased. In conclusion, the incorporation of BNT results in a gradual enlargement and greater uniformity of grain size. And micro-doping of BNT has minimal impact on porosity while aiding in enhancing the density of ceramics.

The PFM technique is utilized to conduct a deeper analysis of the domain configurations in BF-BT-xBNT ceramics with different modifier concentrations. Fig. 5(a–d) shows the vertical PFM information of the ceramics at $x = 0.000$ and 0.001 , including amplitude and phase patterns. It's evident that the amount of BNT doping significantly impacts the size and shape of the domains in the ceramics. The ceramic sample at $x = 0.000$ exhibits heterogeneous domain compositions characterized by macro-scale long stripe domains, leading to an effective piezoelectric response to the tip voltage [15,39]. However, the ceramic sample at $x = 0.001$ exhibits larger island-like domain regions and a few smaller domain regions, indicative of the T phase featuring a singular XRD peak and dielectric relaxation. The incorporation of BNT enhances particles expansion and the formation of microdomains, destroys the original regularity of domain morphology and size to a certain extent, and weakens the ferroelectric long-range order. However, higher phase-contrast is obtained compared to that of the undoped material, which may be one of the sources of high voltage properties [33]. In contrast to the $x = 0.000$ samples, the $x = 0.001$ samples exhibit larger domain structures and lower domain walls, contributing to the domain switching. The complicated domain configuration is observed from the corresponding piezo response image at $x = 0.001$, revealing submicron-size domain morphology. This is because the incorporation of BNT may promote the development of microdomains, and the coexistence of macrodomain and microdomain helps to maintain their rotated states, resulting in a strong piezoelectric response [39]. Furthermore, in comparison to the undoped sample, the major domain regions in the $x = 0.001$ sample are oriented with upward polarization, allowing the polarization vector to easily align in the out-of-plane direction under external stimuli. This may contribute to the enhanced piezoelectric response of BNT-doped ceramics [55]. Therefore, both the quality and quantity of the domains at $x = 0.001$ are significantly improved compared to the undoped ones, leading to a robust

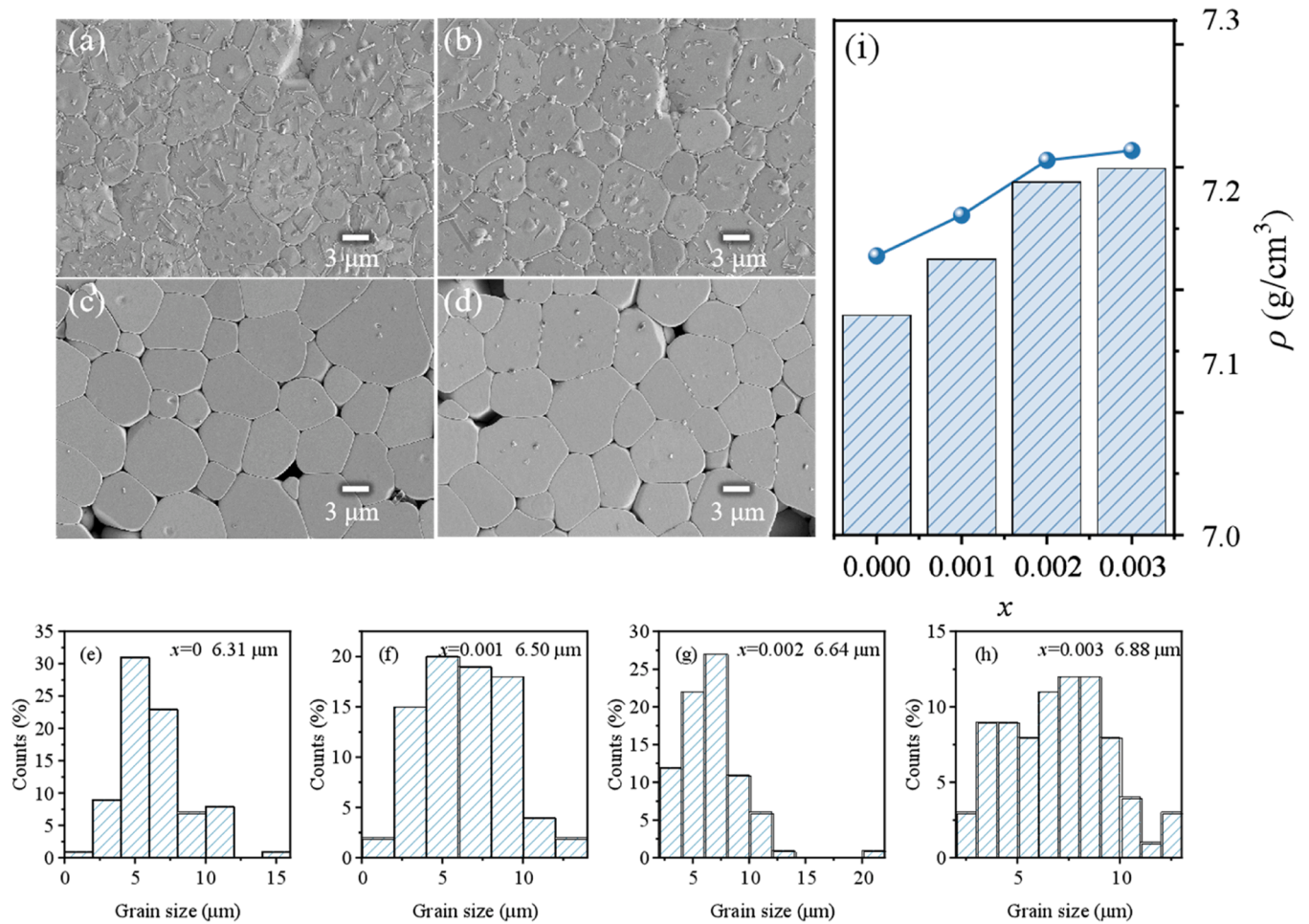


Fig. 4. (a–d) SEM surface micrographs of the BF-BT-xBNT ceramics with $x = 0.000, 0.001, 0.002, 0.003$; (e–h) grain size distributions of the BF-BT-xBNT ceramics with $x = 0.000, 0.001, 0.002, 0.003$; (i) composition dependence of density.

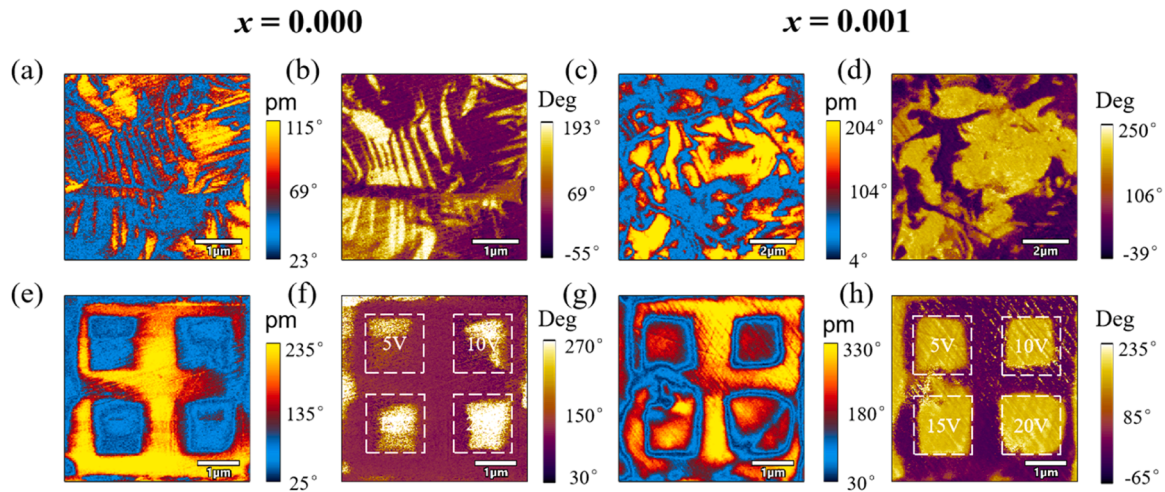


Fig. 5. Vertical piezoelectric force microscopy (VPFM) images information of ceramic with $x = 0.000$ and $x = 0.001$: (a) and (c) amplitude patterns; (b) and (d) phase patterns; (e) and (g) Litho-mode amplitude patterns; (f) and (h) Litho-mode phase patterns.

piezoelectric response, which is consistently linked with increased piezoelectricity. The local poling treatment (litho-mode) of PFM are shown in Fig. 5(e–h). The intensity of the piezoelectric response in the specified region can be reflected by the brightness and size of the domains in the region. At $x = 0.000$, only a limited number of domains

switch at 10 V, while most switch at 20 V. In contrast, for the $x = 0.001$ sample, domains switch completely at a low electric voltage of 5 V, while the rotated domain region at 15 V extends beyond the area indicated by white dashed line square. It can be observed that the increase of polarization voltage causes the intensification of magnetic domain

rotation, denoting that a higher polarization voltage within a specific range help to enhance the piezoelectric properties of ceramics [56]. Meanwhile, BNT-doped samples are more prone to domain switching, implying that BNT doping is beneficial to reduce the domain wall and increase its mobility. These findings suggest that domains in the $x = 0.001$ ceramic are more easily switchable than those in the $x = 0.000$ ceramic, which is one of key factor contributing to the enhanced electromechanical performance observed in the ceramic with $x = 0.001$ [57].

Fig. 6(a₁–a₄) show polarization-electric field (P - E) loops of BF-BT- x BNT ceramics measured at 60 kV/cm from room temperature to 120 °C. It is obvious that the P - E loops of all ceramics demonstrate a noticeable trend of narrowing and elongation as measuring temperature increases, indicating that the remanent polarization (P_r) increase with temperature, while the coercive field (E_c) decrease as temperature rises, as shown in Fig. 6(c) and (d). Typically, the remanent polarization (P_r) of ferroelectric materials decreases with increasing temperature [58]. The anomalous behavior observed in BF-BT- x BNT ceramics may be related to the incomplete saturation of the ceramic's polarization. As the temperature rises, the domain wall pinning effect weakens under the influence of thermal activation energy, the domain wall energy barrier decreases, and domain flipping is enhanced. As a result, an anomalous increase in P_r occurs within a certain temperature range. Additionally, studies have shown that this anomalous polarization behavior is associated with the weakening of antiferroelectric distortion [39] and re-entrant relaxation behavior in BFO ceramics [59]. At 120 °C, P_r decreases with x increasing, and E_c increases with x increasing. This may be related to the increase of lattice distortion caused by BNT doping. Lattice distortion may affect the ferroelectric domain structure in ceramics, making it more difficult to flip polarization, resulting in an increase in P_r and E_c . Fig. 6(b₁–b₄) exhibit bipolar strain loops recorded at

various temperatures from 30 °C to 120 °C. As we can see that positive strain increases with increasing temperatures, attributable to enhanced domain switching facilitated by temperature. Additionally, at high temperature, the strain values of ceramics initially rise and then decline with varying x values, peaking at $x = 0.001$. This phenomenon indicate that BNT doping may enhance improving the ferroelectric properties of the ceramics at high temperature. Fig. 6(e) exhibits unipolar strain loops measured at different temperatures from 30 °C to 120 °C. The results show that the unipolar strain of $x = 0.001$ exhibits a gradual increase as temperature rises, and maintains a relatively consistent growth rate. This occurs because high temperatures furnish sufficient energy to surmount the domain pinning, allowing micro domains to dominate the domain transition, thereby improving the strain response [42]. It is noteworthy that the composition dependence of P_r and E_c value appears abnormal above 100 °C. In addition to the lattice distortion mentioned earlier, PFM results are also consistent with this phenomenon. Under the same electric field and high temperature, the larger macroscopic domain has good stability and still maintains high polarization after the removal of the electric field, while the smaller microscopic domain has poor stability and will return to the polar cluster state after the removal of the electric field [15]. For the sample with $x = 0.000$, macro domains featuring long stripes dominate, leading to a sustained high growth rate of P_r even under elevated temperatures. Conversely, a trace number of nanodomains in the sample with $x = 0.001$ results in a slower growth rate of P_r at high temperature. Therefore, the anomalous behavior of composition-dependent P_r value above room temperature is elucidated.

Fig. 7(a) and (b) show the influence of BNT on the piezoelectric constant d_{33} , planar electromechanical coupling coefficient k_p , dielectric constant ϵ_r , dielectric loss $\tan\delta$ and quality factor Q_m values of x on the BF-BT- x BNT ceramics, measured at room temperature after 24 h of poling. Fig. 7(a) displays an initial increase in the values of k_p and d_{33} ,

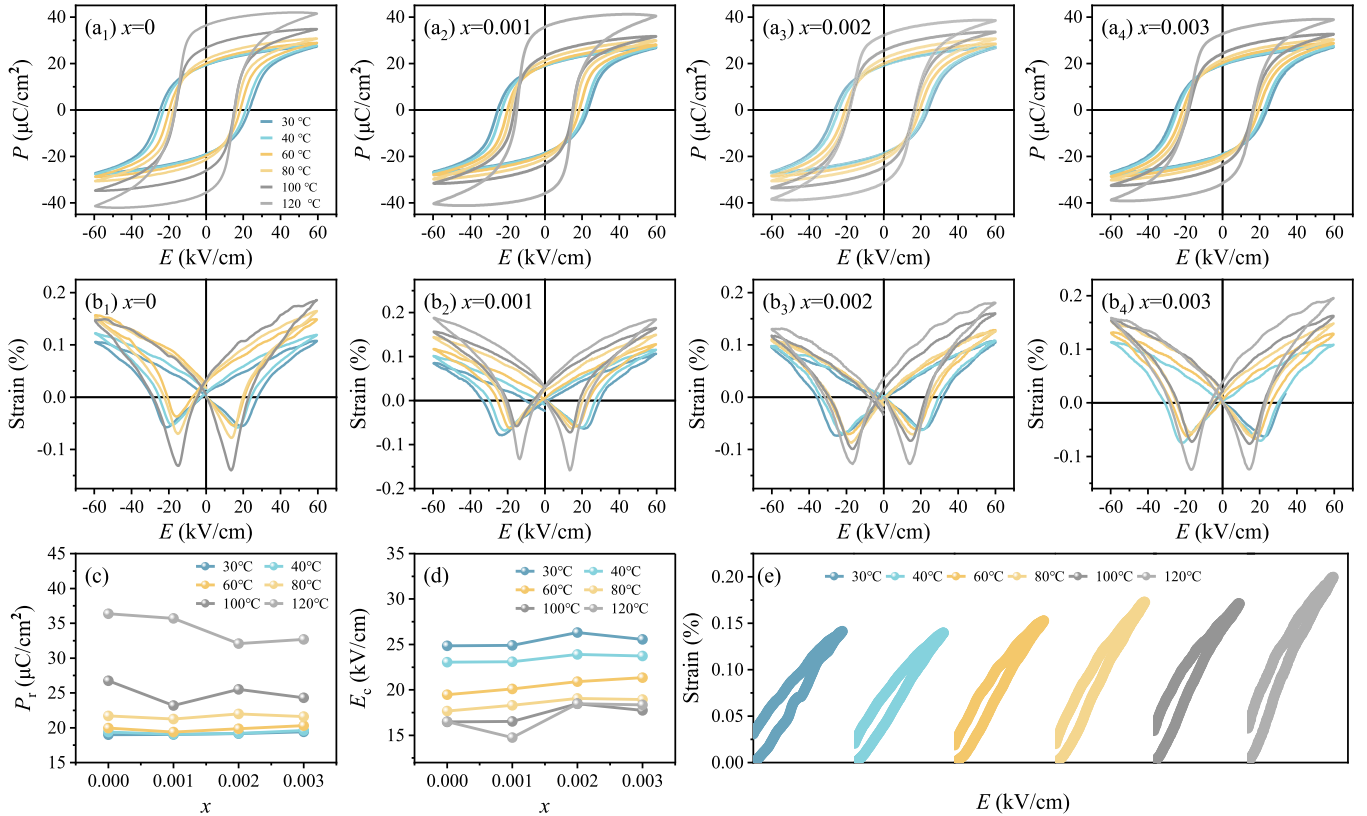


Fig. 6. The temperature-dependent P - E hysteresis loops of the BF-BT- x BNT ceramics at (a₁) $x = 0.000$, (a₂) $x = 0.001$, (a₃) $x = 0.002$, (a₄) $x = 0.003$; Temperature-dependent bipolar strain-electric field (S - E) curves of the BF-BT- x BNT ceramics at (b₁) $x = 0.000$, (b₂) $x = 0.001$, (b₃) $x = 0.002$, (b₄) $x = 0.003$; (c-d) the composition dependence of remanent polarization (P_r) and coercive field (E_c) measured at 30 °C-120 °C; (e) Temperature-dependent unipolar strains loops of BF-BT- x BNT ceramics at $x = 0.001$.

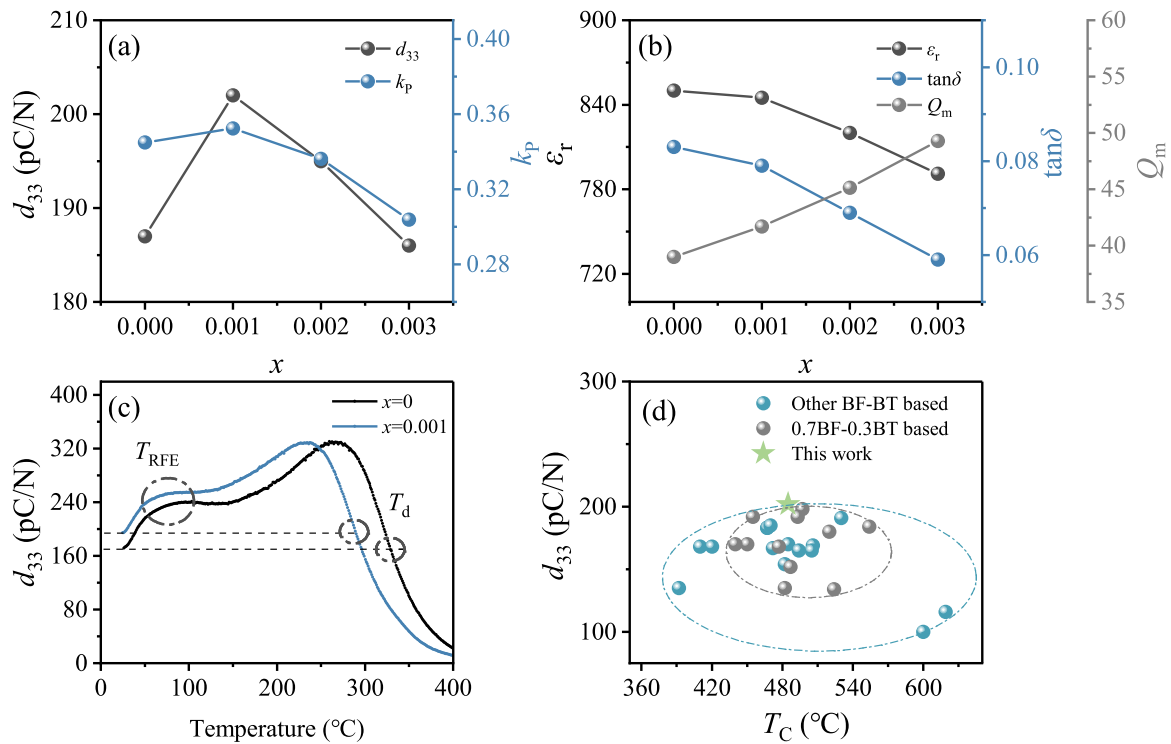


Fig. 7. (a) The composition dependence of electromechanical coupling coefficient k_p and piezoelectric constant d_{33} ; (b) the composition dependence of dielectric loss $\tan\delta$, dielectric constant ϵ_r and Q_m at 1k Hz; (c) the temperature dependent of *in-situ* d_{33} ; (d) comparison of piezoelectric properties.

subsequently followed by a decline, reaching an optimal value near $x = 0.001$ ($d_{33} = 202$ pC/N). In general, the piezoelectric properties of ferroelectrics are assessed through the following Eq. (4):

$$d_{33} = 2Q\epsilon_r P_r \quad (4)$$

Where Q is the temperature-independent electrostriction coefficient, ϵ_r is the dielectric constant, and P_r is the remanent polarization intensity. For low-doping ceramics, Q is almost a constant. Therefore, the increase of d_{33} in the first stage may be influenced by the increase of P_r , and the subsequent decrease is due to the more drastic decrease of ϵ_r [60]. The BNT doping can cause the change of phase structure, resulting in decreased R/T phase ratio, which contributes to the improvement of piezoelectric properties. The dielectric loss $\tan\delta$ and dielectric constant ϵ_r decrease, and the mechanical quality factor Q_m increase with the rise in x , as shown in Fig. 7(b). The reduction of $\tan\delta$ can stem from the higher density and the decreased $V_O^{\bullet\bullet}$ caused by doping, which reduces the leakage current [61]. Moreover, the substitution of Bi^{3+} with BN^{2+} demonstrates acceptor doping, which reinforces domain walls, consequently decreasing $\tan\delta$ within a specified scope. Additionally, the decreased ϵ_r is associated with the lowered R/T phase ratio and the presence of defects, such as $V_O^{\bullet\bullet}$. As illustrated in Fig. 7(c), to further explore the relationship between the piezoelectric properties and temperature of BF-BT-xBNT ceramics, *in-situ* d_{33} values were measured across annealing temperature range from room temperature to 450 °C. As observed, the value of *in-situ* d_{33} increases to T_{RFE} (~80 °C) rapidly and then enters a region of temperature variation with slow growth. This has been shown in the literature to correspond to the re-entrant relaxation temperature, at which the ferroelectric phase undergoes a phase transition to the relaxor phase [59,60]. When the temperature exceeds T_{RFE} , the ferroelectric phase transforms into the relaxor phase, and the increased disorder leads to a reduction in the piezoelectric response. The depolarizing temperature T_d is lower than the phase transition temperature T_m . Additionally, BNT-doped ceramics has a lower T_d than that of undoped ceramics, indicating that doping of BNT may damage the thermal stability of ceramics. Fig. 7(d) compares the d_{33} and T_m of the

ceramics prepared in this work with those of representative BF-BT-based ceramics reported in the literature [3,30,46,62–66]. The introduction of dopants can increase the piezoelectric constant but simultaneously reduce T_m . Conversely, the BF-rich BF-BT-based ceramics exhibit high T_m but limited d_{33} . However, practical ceramic applications demand good comprehensive electrical properties, that is, ceramics with both excellent d_{33} and T_m . In this study, the high values of both d_{33} and T_m suggest that the addition of trace amounts of BNT is an effective method for achieving high overall performance.

4. Conclusion

In order to study and obtain high-performance BF-based piezoelectric ceramics, $(1-x)(0.7\text{BF}-0.3\text{BT})\text{-xBNT}$ ceramics were synthesized using traditional solid phase method, and had a large and stable piezoelectric constant d_{33} in a wide temperature range of 25–300 °C. The peak performance is arrived at $x = 0.001$ ($d_{33} = 202$ pC/N, $T_m = 484$ °C, $T_f = 466$ °C, $k_p = 35$ %, $P_r = 19.02$ $\mu\text{C}/\text{cm}^2$ and $Q_m = 41.7$). All ceramics exhibit a biphasic coexistence of both R and T phase, with the proportion of R to T gradually decreasing as the BNT content increases, accompanied by the presence of relaxation properties. Furthermore, a small amount of BNT doping induces octahedral distortion and lattice anisotropy, and the competitive interaction between the increased distortion and the decreased trigonal phase content would lead to the stabilization of T_m . Meanwhile, the synergy effect of reduced R/T ratio, appropriate grain size, enhanced relative density and diminished leakage current contributes to favorable ferroelectric and piezoelectric properties. Moreover, BNT doping induced an enlargement of average grain size and the appearance of refined grains, which may promote the coexistence of macroscopic and nano-magnetic domains. This phenomenon enhances domain wall mobility and switching behavior, synergistically resulting in a crucial improvement in piezoelectricity. Consequently, the BF-BT-xBNT ceramics is considered as a potential candidate material for high temperature applications.

CRediT authorship contribution statement

Zhongqin Yang: Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mingyue Mo:** Writing – review & editing, Methodology, Conceptualization. **Ning Chen:** Writing – review & editing, Data curation. **Manjing Tang:** Writing – review & editing, Data curation. **Zhi Tan:** Writing – review & editing, Data curation. **Jie Xing:** Writing – review & editing, Data curation. **Jianguo Zhu:** Writing – review & editing, Resources, Funding acquisition, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by the National Natural Science Foundation of China (52032007), Sichuan Science and Technology Program (2023YFG0042), the Natural Science Foundation of Sichuan Province (24NSFSC5385), and the Fundamental Research Funds for the Central Universities. We also appreciate Ms. Hui Wang from the Analytical & Testing Center of Sichuan University for SEM characterization.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jallcom.2024.178322](https://doi.org/10.1016/j.jallcom.2024.178322).

Data availability

Data will be made available on request.

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